Arista Access Node MAR-ACC-36 Optical Transport Failure SOP (ALARM_5916)

Node: MAR-ACC-36 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_5916

Description

The failure of Optical Transport on Arista Access Node MAR-ACC-36 (Role: Access, Vendor: Arista, Node ID: MAR-ACC-36) is attributed to the degradation of SFP/XFP modules, fiber cuts, or Rx/Tx power loss, resulting in a warning alarm (ALARM_5916). This phenomenon can be ascribed to the inherent characteristics of optical transmission systems, where signal integrity heavily relies on the physical properties of the fiber medium and the transceivers. The optical signals are prone to dispersion, attenuation, and other impairments that can be exacerbated by environmental factors, such as temperature fluctuations, vibration, or electromagnetic interference. Furthermore, the SFP/XFP modules, being the interface between the optical signals and the Arista equipment, can experience degradation due to wear and tear, excessive power consumption, or manufacturing defects.

Pre-requisite Checks

- Verify the site location and access to Marathahalli Core. Ensure remote hands are dispatched as necessary.

- Check the status of redundant links to determine if the failure has impacted network connectivity.

- Review the event history to confirm the occurrence of similar events and assess the root cause.

Potential Root Causes

Cause	Probability
SFP/XFP module degradation due to excessive power consumption or manufacturing defects	High
Fiber cuts or breaks due to physical damage or environmental factors	Medium
Rx/Tx power loss due to signal degradation or transceiver issues	Low

Diagnostic Commands & Expected Outputs

```
Command: show interfaces xe-0/0/0 detail, Expected Output: Interface xe-0/0/0, Enabled, Physical link is Up, Description: This command retrieves detailed information about the xe-0/0/0 interface, including its link status, speed, and duplex mode. The expected output indicates that the link is up and functioning, ruling out physical link failures as the root cause.

Command: show log messages | match ALARM_5916, Expected Output: ALARM_5916: Warning: Optical Transport Failure on interface xe-0/0/0, Description: This command searches the system log for messages related to ALARM_5916 and displays the exact alarm message. The expected output confirms the occurrence of the alarm and provides insight into the affected interface.
```

```
Command: show hardware internal optics | match xe-0/0/0, Expected Output: Optics type: SFP+, Transceiver serial number: ABC12345, Description: This command retrieves information about the optics module installed on the xe-0/0/0 interface, including its type and serial number. The expected output enables further investigation into the optics module's condition and potential issues.
```

Resolution Steps

Step 1: Step: Replace the SFP/XFP module with a new one, Command: replace optics xe-0/0/0 SFP+-ABC12345, Description: This step involves replacing the degraded SFP/XFP module with a new one, ensuring that the transceiver is compatible with the Arista equipment and the existing fiber infrastructure.

Step 2: Step: Verify fiber integrity and repair or replace the affected fiber, if necessary, Command: no shutdown, Description: This step involves checking the fiber's physical condition and repairing or replacing it, if damaged. The no shutdown command is used to enable the interface after the fiber has been repaired or replaced.

Step 3: Step: Adjust the Rx/Tx power levels to optimize signal integrity, Command: set interfaces xe-0/0/0 rx-power auto, Description: This step involves adjusting the receive and transmit power levels on the xe-0/0/0 interface to optimize signal integrity and ensure reliable transmission.

Post-Check Verification

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Command: show interfaces xe-0/0/0 detail, Expected Output: Interface xe-0/0/0, Enabled, Physical link is Up, Description: This command verifies that the link is up and functioning after the resolution steps have been executed.

Command: show log messages | match ALARM_5916, Expected Output: ALARM_5916: Warning: Optical Transport Failure on interface xe-0/0/0 (resolved), Description: This command verifies that the alarm has been resolved and the system log reflects the corrected status.

Command: show hardware internal optics | match xe-0/0/0, Expected Output: Optics type: SFP+, Transceiver serial number: DEF67890, Description: This command verifies that the replaced optics module is correctly installed and functioning.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP module, if replaced, Command: replace optics xe-0/0/0 SFP+-ABC12345, Description: This step involves restoring the original SFP/XFP module, if replaced, to its original configuration.

- Step: Verify fiber integrity and repair or replace the affected fiber, if necessary, Command: no shutdown, Description: This step involves checking the fiber's physical condition and repairing or replacing it, if damaged. The no shutdown command is used to enable the interface after the fiber has been repaired or replaced.

- Step: Adjust the Rx/Tx power levels to optimize signal integrity, Command: set interfaces xe-0/0/0 rx-power auto, Description: This step involves adjusting the receive and transmit power levels on the xe-0/0/0 interface to optimize signal integrity and ensure reliable transmission.